Japan Train Coverage

Identifying underserved areas by Japanese train stations

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ABSTRACT

The purpose of this project is to help identify areas in Japan that are underserved or overloaded based on the relationship between train station coverage and population density. The goal is to identify areas that do not have an appropriate number of stations, stations in wrong relative location for population centers, or general outliers. It may provide insight into where additional train stations may be beneficial in the future and help provide public transport to those areas.

This analysis may be performed through several methods either by dividing up the country based on prefectures, grids, or based on the areas surrounding the train stations themselves. This may provide different points of view to identify such underserved areas. Finally, analysis of these outliers may be performed not just through GIS mapping methods but using cluster analysis to identify outliers based on population or identify the common relationship of population to number of stations.

Through the process of finishing the project, the focus of the project changed to focus more on the change of population over time, and its effect of population on train station closures or additions in each area.

Introduction

This project attempts to fix the problem of easily mapping and identifying areas that are missing train station coverage. If any areas are identified, such underserved areas, this relation may have other indicators such as transportation problems, excessive car usage, or lack of public transport options.

Additionally, if any areas are lacking in public transport, this may provide transportation issues for foreign tourists. Such tourists may lack the ability to get a driver's license, limiting travel outside of city centers. Finally, a lack of knowledge about other methods of travel may be present in areas where public transport options such as trains or buses are lacking.

Related Work

Several solutions and research papers currently exist which primary research the relationship between train lines or stations, and the relationship to population directly around those stations. One such project by Yuka Otsuga [1] reviews the relationship between these stations, population density, and specific JR rail lines. Additional projects exist which review the effect of high-speed rail and how high-speed rail affects population density or locations around the country [2].

These papers provide extensive details around how train stations and high-speed public transport affect the population of Japan, and its population centers. Ideally, this project aims to invert such analysis by performing gap analysis on areas which are underserved by public transport options including trains.

Proposed Work

The main tasks of the proposed work are to identify statistical analysis on the relationship between population density and train station coverage. Additionally, visualization of this relationship is core to this project and will be visualized by overlaying the data on a map of the area. Temporal or historical changes may be investigated depending on the availability of the data.

In addition to mapping the relationship on the map, additional visualizations will be created to identify relationships and identify populations which are outliers in the station per-capita relationship. This will use clustering to identify such outliers, including extreme station per-capita values both positive and negative relationships. This analysis will be a clustering problem to identify outliers; however, it may change depending on additional insights during data discovery.

This project will primarily use a Jupyter notebook to query the specific APIs and store the data in parquet files on the local system. Once the API data has been downloaded analysis will be done in Python to query the files and perform analysis using libraries such as GeoPandas. All analysis will be performed and stored locally.

Evaluation

Evaluation of this analysis will likely take the form of predicting population changes and per-capita station requirements of a given location based off the existing locations. Such evaluation could include predicting population change and determining the required increase of train availability. This analysis will depend

[1] [1] Otsuga, Yuka, et al. "Relational Analysis of User Density, Number of Train Services and Population around Rail Stations on Local Railways in Japan." Transportation Research Procedia, vol. 82, 2025, pp. 2137–2148, www.sciencedirect.com/science/article/pii/S2352146524004769, https://doi.org/10.1016/j.trpro.2024.12.177. Accessed 28 Jan. 2026.

[2] Wang, Jingyuan, et al. "Do High-Speed Railways Have an Impact on Population Change? Evidence from Japan." Transportation Research Part a Policy and Practice, vol. 187, 1 Aug. 2024, pp. 104187–104187, https://doi.org/10.1016/j.tra.2024.104187.

on the historical change of data sources and time constraints of this project.

One ideal evaluation metric will be to identify outliers during clustering and finding any areas which are underserved. This metric will be expanded during data exploration. Additionally, data sets will likely not be split, due to the limited access to data. If project time restrictions allow, different datasets will be created for different years or changes over time.

No throughput concerns are currently present, as API queries will download a stations list, download population density maps offline, and analysis will be performed offline.

Discussion

The progression of this project at the time of this writing has already begun. Initial research on data sources, mapping techniques, and ways to split the country has already been investigated.

The timeline going forward for the remainers of this class will be roughly to finish one task per week as follows:

- Week 1: Train station mapping
- Week 2: Population density mapping
- Week 3: Relationships between population and stations
- Week 4: Pattern analysis and clustering
- Week 5: Project wrap up and documentation.

Potential challenges and limitations of this project are the density of the population maps. While population maps have been identified, the resolution of these maps will determine how big or small an area that this analysis will be over, such as on a prefecture analysis or a country wide analysis.

Depending on the relationships identified or lack of resolution on the population maps, backup analysis such as environmental factors may be performed to identify clusters or common patterns.

This project helped visualize Japan's overall population loss, and how such population loss affects certain areas and its resulting train station closures. One major lesson learned through this project is that this analysis requires a larger timeframe of analysis such as changes over decades rather than a limited glimpse of 2016 to 2025 in this project.

Interim Assessment - 2/11/2026

This section will provide an overview of the status of the project and findings found since beginning the project. Overall, at the point of the project, the project is ahead of schedule and will be completed on time.

As of the time of this project all data has been obtained, except for periodic changes in train stations over a decade. This data is available; however, it is currently not returning data properly. Currently all data for the following areas has been obtained, and imported into the data pipeline:

- Train station locations – *Overpass*[3]
- Population density maps - *WorldPop*
    - 1km resolution
    - Years 2016[4] and 2025[5]

To aid in preprocessing the data before performing analysis of the relationships between population and station locations, multiple data normalization processes and organization occurred. First, stations were simply queried for latitude and longitude, to allow for mapping of the stations. Next population data from 2016 and 2025 are read in from raster files (Tif) which allow for detailed representation of the population distribution of Japan. This initially began with a raster file with 21x30m resolution but was too large for timely processing. This changed to 1km resolution of the population data, which is now being used for both years' analysis.

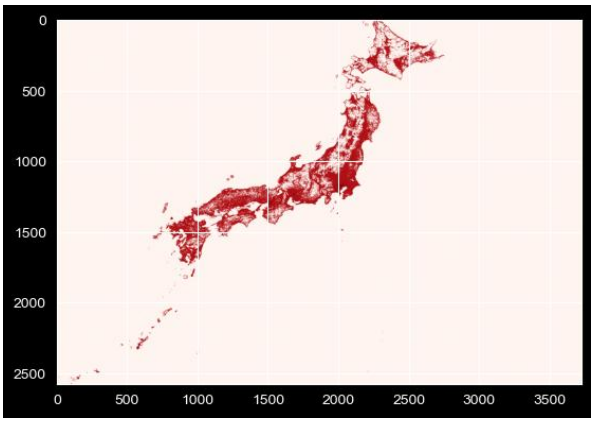

Figure 1: Population Raster

Once the population and station lists had been generated, Japan was divided into H3 hex grids to allow for analysis on a subsect of Japan at a time. Each one of these grids are analyzed at a time; by using attributes such as id, centroid of the grid, population subsect, and train station count within the specific grid. The population map from the raster was clipped to obtain the population within a specific grid. This clipping was performed using rasterio to mask the population count within a clipped H3 grid coordinate.

[3] https://wiki.openstreetmap.org/wiki/Overpass_API
[4] https://hub.worldpop.org/geodata/summary?id=77814
[5] https://hub.worldpop.org/geodata/summary?id=77823

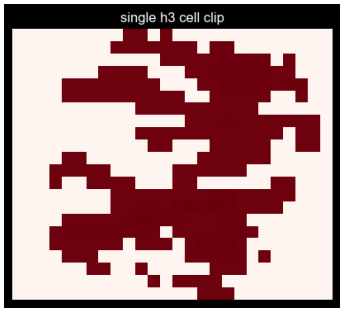

Figure 2: H3 Population Mask

To aid in organizing this analysis, a Python class was created to track the individual grid and the following attributes. One object of this class is created for each grid created during the process. These attributes are focused on the centroid or center of the H3 hex grid as a reference point.

- Id
- Centroid (Latitude, Longitude)
- Distance from centroid to nearest station
- Population
- Station count
- Station list

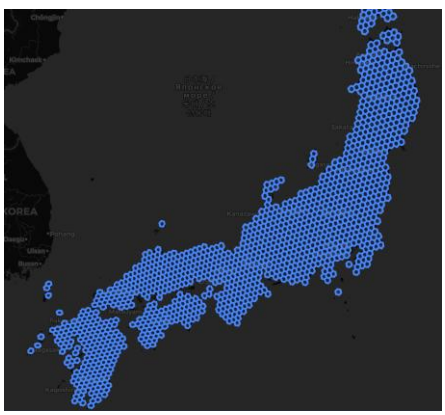

Figure 3: Japan H3 Grid Division

From there stations are assigned to a grid based on their location. While this query has been changed to a GeoPandas spatial join, the loop below shows the same effect.

```
Station: 巻向 found in grid: 1348
Station: 三輪 found in grid: 1348
Station: 香久山 found in grid: 1348
Station: 大和朝倉 found in grid: 1348
Station: 長谷寺 found in grid: 1348
Station: 耳成 found in grid: 1348
Station: 刺巻 found in grid: 1349
Station: 田沢湖 found in grid: 1349
Station: 羽後中里 found in grid: 1349
Station: 松葉 found in grid: 1349
Station: 羽後長戸呂 found in grid: 1349
```

Figure 4: Station to Grid Assignment

Currently data warehousing is stored in a series of Parquet files primarily around the station list (*station-list-YEAR.parquet*) and a final generated parquet file (*final-station-table.parquet*). This final table creates an index of the grid attributes as listed above, which helps track all related fields and relationships. All data is stored in a Pandas dataframe, before being written to the final parquet file.

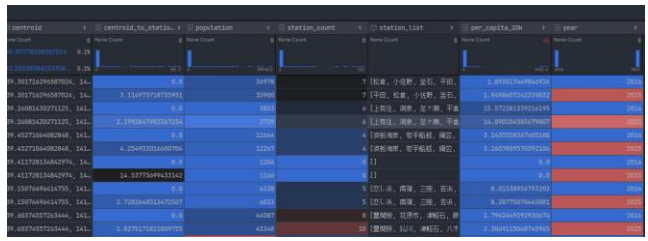

Figure 5: Grid Data Sample

While currently the data is stored in those parquet files, ideally after the class ends, this project will be converted to tables in Databricks to allow for use of Spark and faster processing methods. This conversion from parquet to delta tables should be possible outside of the time restraints of this class. However, I do not believe additional tables at this point are necessary besides the two tables. Additionally, due to the size of this project, helpers and API scripts may need to be separated out of the notebook into separate helper scripts and table creations.

As the project progressed, parquet usage did get convoluted, the move to Databricks or a SQL database will provide additional clarity and easier management of data. Editing these parquets and dataframes became more complicated to manage as the project progressed.

Once the data has been created this allows for analysis between the population of a grid, number of stations, stations per capita, and eventually how stations change over time based on population or vice versa. These metrics such as population change, station count change, and others are then used for modeling such as clustering, finding anomalies, and hopefully identifying patterns in the future based on population change. This is not currently working at the time of the report, such data is available, so time permitting this will be completed before the class ends.

Currently manual data modeling such as clustering has been performed to include K-Means to identify outliers within the data relationships. This analysis can help identify stations with a low station count per capita and help identify outliers. This per capita relationship is already in the analysis and helps provide more accurate relationships. This analysis is aided by using existing tools including Scikit-Learn's K-means and other open-source Python libraries.

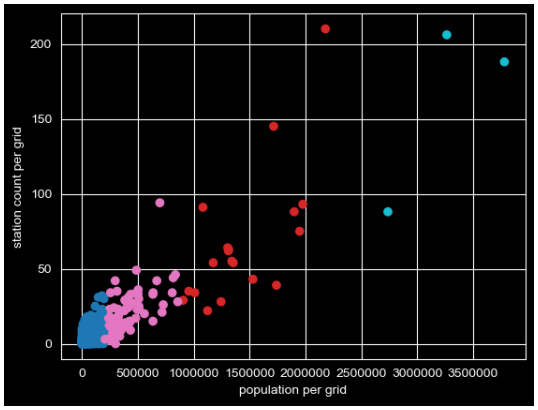

Figure 6: Population vs Station Count

Preliminary results have identified several stations which have a larger per-capita relationship, such as stations with more stations that are typically required. Additionally, several grids were identified having larger distances to stations than typical.

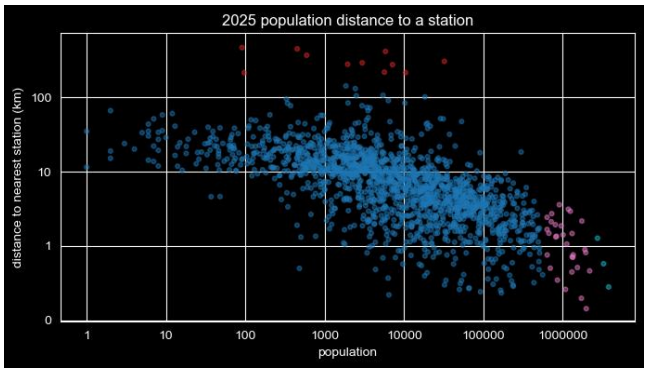

Figure 7: Population vs Distance to Station

Analysis such as classification is additionally in progress to determine if a grid is in a rural or city center to provide easy classification of a grid and therefore give more context to its attributes. This will be a binary attribute within the grid class.

In addition to standard data modeling based on the attributes generated through processing, heatmaps and other relationships provide insight into the spread of stations throughout the country. This mapping is done with prebuilt tools such as Folium and GeoPandas.

Figure 8: Station Distribution Heatmap

As mentioned previously, time permitting, additional analysis and predictions will be added if previous year train station counts can be added to the dataset. This will allow for deeper relationships between population growth based on station additions in rural areas.

At the time of this report update there are several issues within the pipeline, that while not currently providing breaking errors, can allow for slowdowns or data misinterpretation.

Regarding the Overpass API for listing the train stations, several issues currently exist within the project. First, the query of pulling train stations was limited to 2000 stations as it relied on user flagging of stations, an updated query was added to pull objects which are public transport stations and are also serving trains, which resulted in a more accurate 8500 stations. This data was more in line with the accurate count of all train stations in Japan. Additionally, as mentioned before, pulling historical data should be possible through the API, however, is not currently working at the time of this report.

Second, several issues were identified during defining the resolution of the H3 hex grids, whether to make the grids a smaller area or a larger area. If a small area was defined, this would allow for high resolution analysis of an area and additionally would limit the clipping of cities directly on the coast or sharp curves. However, with a smaller area these blocks would not provide a good distribution of locations and population as it was too granular. Additionally, this division would take longer on the execution time. A large area would allow for more representative areas, however, would result in large swaths of the coast being missed due to the hex division. Due to both, a medium resolution has been set on the number of grids, resulting in roughly 1700 different grids being analyzed.

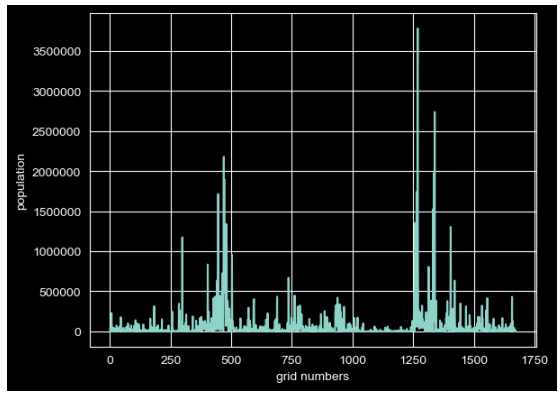

Figure 9: Grid Population Distribution

Additionally, validation of the data has been spot checked by checking the accuracy of the population count for a grid in tools such as QGIS. Several grids have been validated as accurate; however, more testing and asserts are needed to ensure that the population count, and station assignments are valid.

In conclusion, at the time of this status update, the project in on track as intended. Time limitations may limit the ability to perform temporal analysis through changes in population, changes in train station coverage, and how stations affect population in

rural areas. This is intended analysis, but time limitations do currently exist for the class.

Interim Assessment - 2/18/2026

The second phase of the project focused heavily on the historical differences between 2016 and 2025 by comparing population and train station dataset differences over these years.

Early in the second phase of this project, the Overpass API was removed due to API limitations of historical data, in favor of snapshot files from OpenStreetMap's PBF files. These files provided train station data for a given year, allowed for more complicated queries, and allowed for historical comparison of train stations between 2016 and 2025 for this project.

Population raster data was not affected by this API and was obtained through WorldPop for the years 2016 and 2025 for historical comparison.

As mentioned in previous assessments, numerous outliers were identified in which grids were lacking a train station but were very distant to a neighboring grid that did contain a station. This initially skewed results.

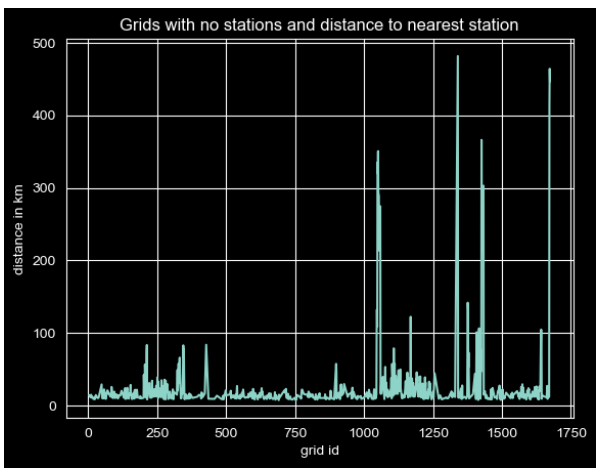

Figure 10: Outliers Identified

Through analysis of these neighbor distances, the outliers were identified primarily as islands which did not have a train system. Larger islands such as Okinawa were not flagged as they do have train systems. Filtering out these points allowed for better analysis of relationships.

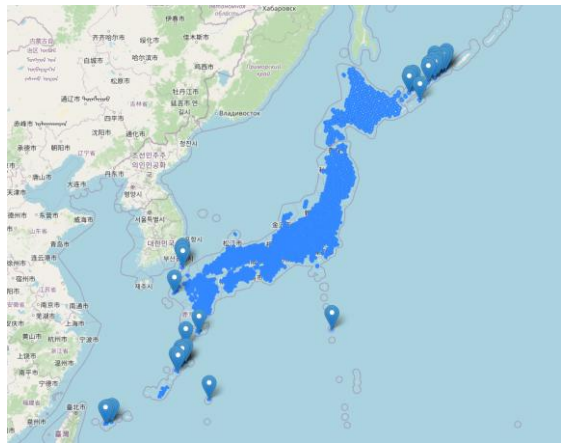

Figure 11: Outlier Mapping

Additionally, the use of WorldPop data allowed population changes to be mapped. This provided several details such as overall population loss, which grids lost population, and where those decreases were affected the most.

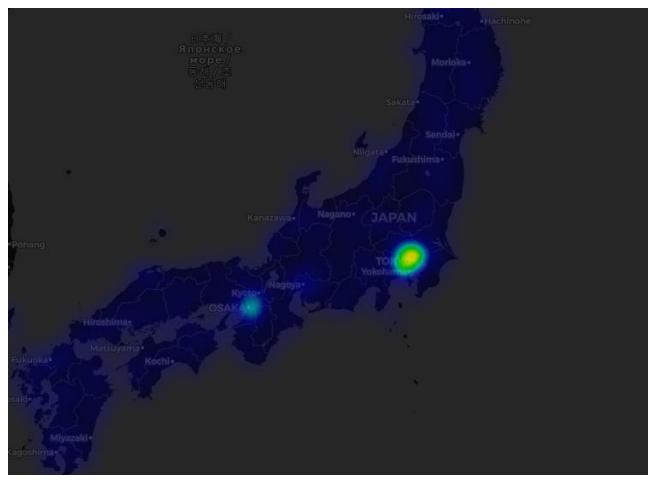

Figure 12: Population Decrease Heatmap

Overall, as seen in the figure above, the country as a whole loss population between 2016 and 2025, with higher losses in urban areas due to population density. Additionally, this change was spurred by the migration of people from rural areas to more urban city centers. This migration of people did affect the closure of stations in some rural regions.

In addition to population differences between those years, station changes were identified over the years between 2016 and 2025. Overall, over 100 net stations were removed in those years. This migration favored removing lines or stations in rural areas, while adding additional stations in more urban areas such as Tokyo.

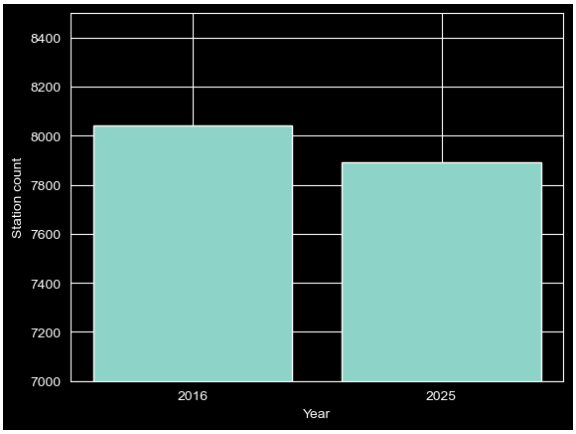

Figure 13: Train Station Count Decrease

One feature this project's analysis provided was historical comparison of train station additions throughout the years. By identifying the difference between a grid's station count, additions or removals of train lines were identified. One such addition was the addition of 4 new train stations in Okinawa, through the expansion of an existing train line. This extension in 2019 introduced the four stations Ishimine, Kyozuka, Urasoe-Maeda, Tedako-Uranishi, which can be seen in the project's data below. This change in stations between the project's 2016 and 2025 range can be validated.

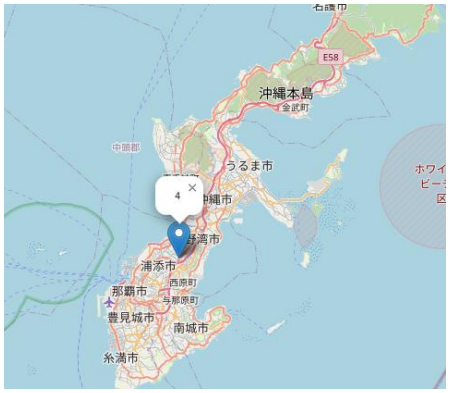

Figure 14: Train Station Count Increase (Okinawa)

This analysis of train station counts was produced on a grid-by-grid basis, allowing for easy identification of big changes in station count. Such exceptions can be seen below in which one line closed more than 20 stations, while additional grids in more urban areas added more, as seen in the Figure below.

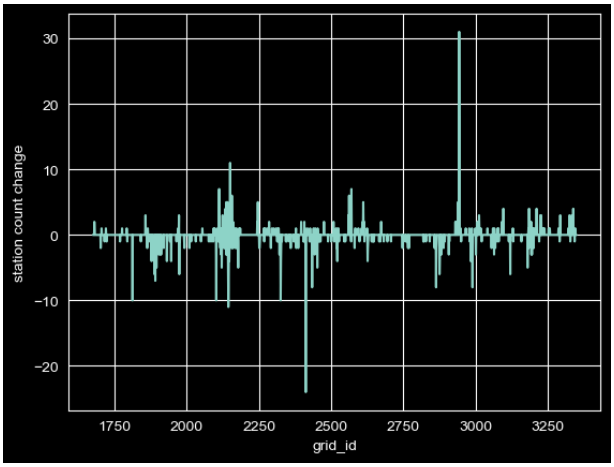

Figure 15: Station Count Change (2016 to 2025) by Grid

One significant outlier identified in this project's analysis above highlights once grid losing over 20 stations in the decade between 2016 and 2025. This loss of stations can be attributed to the closure of the Sanko line[6] near Hiroshima. This line was permanently closed in 2018 due to limited usage of the train line, and as a result closed the stations on the line, in replacement for a bus line. This sharp drop in station count can be seen in the figure above and overall identified in the project.

For this line specially, while there exists a population drop in the area from 2016 to 2025, as seen in the next figure, this sharp drop is not necessarily correlated. While there was a population drop in the area in the decade from 2016 to 2025, the closure of the line was due to declining population in the area over 40 years. As noted in the article, peak usage from 1987 to 2014 was nearly 1/10th of the usage in prior decades.

The figure below shows the population loss for each grid between 2016 and 2025.

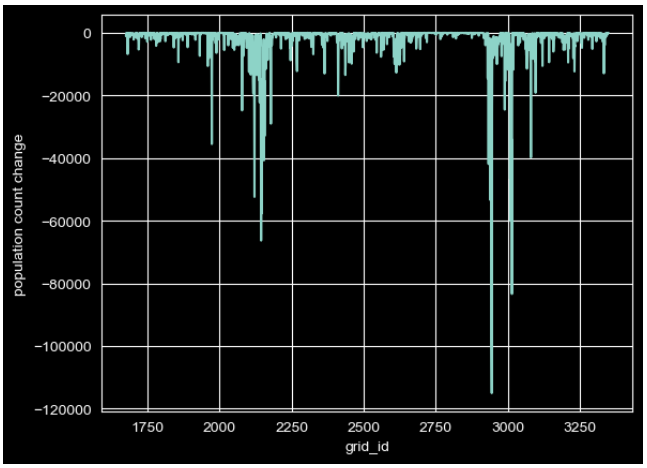

Figure 16: Population Change (2016 to 2025) by Grid

[6] https://en.wikipedia.org/wiki/Sank%C5%8D_Line

This lag in Japan's population decline vs train station closure can span decades. While this is a very good metric to analyze, this is out of scope of the current project. As such, such direct relationships between populations change in each decade, vs train station changes in each decade will not be directly correlated.

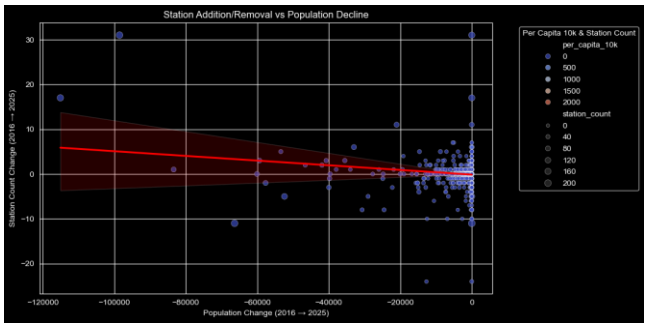

Figure 17: Station Change vs Population

Conclusion

Overall, this project aims to identify areas of Japan that are underserved based on population density, the number of stations, and the location of these train stations. This project will not focus on the other methods of public transport including buses, while solely focusing on trains.

When finished, the project aims to identify zones, prefectures, or population dense areas which are lacking stations or an appropriate amount of the population they aim to serve. This aims to identify any lack of coherent public transportation in those areas. Future work after this should include high resolution population maps if available, in addition to mapping those specific underserved areas based on population change.

As expected, there was a strong correlation between the population of a given grid, and its number of train stations. This proved a correlation that as the population went up the station count for a given grid generally went up. This was not a linear relationship as very urban stations did service a higher number of people for a given station.

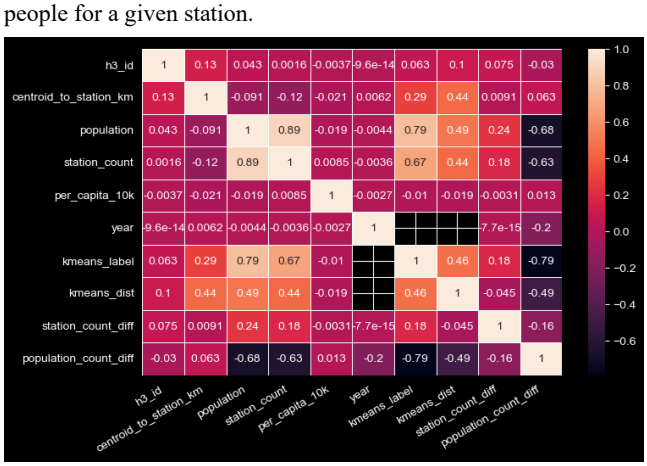

Figure 18: Final Correlation Matrix

One surprising item in the discovery phase was that sharp population changes in each area did not have immediate effects on the station count. As described earlier, a noted 24 station drop wasn't directly in relation to the population change, but its population change in the previous decades leading up. This can cause a bit of a lag between stations. While in that grid, population dropped, it took several decades for the stations to be removed due to limited demand.

Additionally, while the population of Tokyo in general may be declining, additional stations were added to help lower the load on existing stations in those congested areas.

As outlined in the introduction, this project aimed to identify underserved areas by trains, as well as help predict where additional stations may be needed. This project did not achieve both of those initial goals, as such prediction would require more than 10 years of difference, additional decades are needed to see these trends. While those initial objectives were not met, this project allowed for identification of events that happened probably in previous years due to this lag and allowed for quick identification of areas which lost significant train coverage due to both population change and train lines closing.

This project pointed out several key findings throughout the analysis phase including Japan's population loss, migration from rural areas, as well as resulting train closures in rural regions.

As mentioned previously, future work of this project will likely include moving the project to a more sustainable data warehouse, to allow for easier storage and management of data. Additionally, if time permits, additional years will be added to allow for review of changes over several decades instead of years.

REFERENCES

[1] Otsuga, Yuka, et al. "Relational Analysis of User Density, Number of Train Services and Population around Rail Stations on Local Railways in Japan." Transportation Research Procedia, vol. 82, 2025, pp. 2137–2148, www.sciencedirect.com/science/article/pii/S2352146524004769, https://doi.org/10.1016/j.trpro.2024.12.177. Accessed 28 Jan. 2026.

[2] Wang, Jingyuan, et al. "Do High-Speed Railways Have an Impact on PopulationChange? Evidence from Japan." Transportation Research Part a Policy and Practice, vol. 187, 1 Aug. 2024, pp. 104187–104187, https://doi.org/10.1016/j.tra.2024.104187.